

☒ 1. The Samba Server – Networking with Windows Systems

☒ What is Samba?

- Samba is an open-source implementation of **SMB/CIFS** protocol for **Linux/Unix** systems to **share files and printers with Windows clients**.
 - Samba enables **Linux to act as a Windows file server** (like Active Directory domain controller or member).
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☒ Key Features:

- File sharing with Windows (`\\servername\sharename`)
- Printer sharing
-

Integration with Active Directory

- User authentication via local or AD credentials

☒ Installing and Configuring Samba (Ubuntu/Debian):

1.

Install Samba:

```
bash
```

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```
sudo apt install samba
```

2.

Samba Config File: /etc/samba/smb.conf

3.

Create a Shared Directory:

bash

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```
sudo mkdir -p /srv/samba/shared sudo chmod 2775 /srv/samba/shared sudo chown nobody:users /srv/samba/shared
```

4.

Edit smb.conf:

ini

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```
[shared] path = /srv/samba/shared browseable = yes read only = no guest ok = yes
```

5.

Restart Samba:

```
bash
```

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```
sudo systemctl restart smbd
```

6.

Access from Windows:

-

Use \\<Linux-IP>\shared in File Explorer.

☒ Adding Samba Users:

```
bash
```

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```
sudo smbpasswd -a <username>
```

☒ 2. Configuring a DHCP Server

☒ What is DHCP?

- Dynamic Host Configuration Protocol assigns **IP addresses automatically** to devices on a network.
 - Manages IP lease, subnet mask, DNS, gateway, etc.
-

☒ Installing DHCP Server:

On Debian/Ubuntu:

```
bash
```

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```
sudo apt install isc-dhcp-server
```

On RHEL/CentOS:

```
bash
```

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```
sudo yum install dhcp
```

☒ Key DHCP Files:

- Config file: /etc/dhcp/dhcpd.conf
- Lease file: /var/lib/dhcp/dhcpd.leases
- Interface: /etc/default/isc-dhcp-server

☒ Sample dhcpd.conf:

conf

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```
default-lease-time 600; max-lease-time 7200; subnet 192.168.1.0 netmask 255.255.255.0 { range 192.168.1.100 192.168.1.200; option routers 192.168.1.1; option domain-name-servers 8.8.8.8, 1.1.1.1; option domain-name "example.com"; }
```

❏ Start and Enable Service:

```
bash
```

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```
sudo systemctl start isc-dhcp-server sudo systemctl enable isc-dhcp-server
```

❏ 3. Configuring a DNS Server (*Summary*)

- Use BIND9 or dnsmasq

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Primary file: /etc/named.conf or /etc/bind/named.conf.options

- Set up forward and reverse zone files.
- Use named-checkconf and named-checkzone to validate.
- Start DNS service with:

bash

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```
sudo systemctl restart bind9
```